



Research paper

Spatio-temporal graph interaction networks for teacher behavior description in classroom scene

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ABSTRACT

Teacher behavior description is crucial for improving teaching effectiveness and understanding classroom dynamics. Traditional methods rely on meticulous manual observation and recording, which is extremely complex and time-consuming when dealing with many teaching videos. Recently proposed video captioning technologies can automatically analyze and describe object behaviors, reducing manual intervention and providing powerful support for improving teaching effectiveness. Existing video captioning methods, however, fall short in describing teachers' behaviors because they overlook the temporal progression of teachers' actions, the background changes between frames, and the frequent interactions between teachers and students in real classroom environments. To address these issues, we propose a video captioning method leveraging a spatio-temporal graph interaction network (STGIN) to describe teachers' behaviors in classroom scenes. Comprising the teacher–student spatial interaction module (TSI), teacher temporal context module (TTC), and description generator, STGIN captures spatial interaction relationships between teacher and students, extracts temporal dynamics features of the teachers' behaviors, and precisely generates teacher behavior descriptions. To validate the proposed scheme, we collected a teacher behavioral description (TBD) dataset consisting of 2000 videos. Extensive experiments on our TBD dataset and two generic scene datasets confirm the effectiveness and robustness of the proposed STGIN in real classroom environments.

1. Introduction

Teaching behavior analysis is a pivotal avenue for enhancing educators' professional capabilities. Educators can comprehensively understand their teaching dynamics and improve their professional skills by observing, recognizing, and describing teacher behaviors (Nagro and Cornelius, 2013). Early methods of teaching behavior analysis used manual recording and description of teacher behaviors (Dai et al., 2023) to generate teaching evaluations, which strongly depended on the subjective experience of experts (Rodgers et al., 2019). With the widespread adoption of digital classrooms, many teaching videos have been collected from micro-classes and recorded lessons, making manual methods both time-consuming and labor-intensive. Therefore, researchers have begun to explore automated analysis techniques of teacher behaviors through these massive teaching videos. Recently, some methods have attempted to use action recognition techniques (Shiyan et al., 2022; Chen and Guan, 2022) to quantitatively evaluate the collected data, thereby summarizing the types or activity rates of teacher behaviors. While these methods can provide a certain level of objective behavior analysis, they may fail to understand the specific context and meanings behind teachers' behaviors, limiting the

professional development of educators. Recent studies on micro-action recognition (Guo et al., 2024; Li et al., 2025a) further reveal that subtle, low-intensity movements can reflect teachers' psychological states and intentions, which are often overlooked by conventional action recognition frameworks. This highlights the need for fine-grained behavioral modeling in educational contexts.

Compared to the quantitative evaluation of action recognition technologies, the description of teaching behaviors can document and portray the specific actions of teachers during the instructional process and the surrounding context (French and Galloway, 1968). This approach can provide the underlying intentions and educational objectives behind teachers' actions, which helps to promote teaching reflection and professional development of teachers. Existing methods of teaching behavior description, however, remain at the manual annotation stage and quickly and objectively process a large amount of instructional data. The recently proposed video captioning technology can automatically describe the behaviors of individuals in videos, providing a potential solution for automatically generating teaching behavior descriptions in educational videos. Furthermore, video captioning technology can process large amounts of video data in real time

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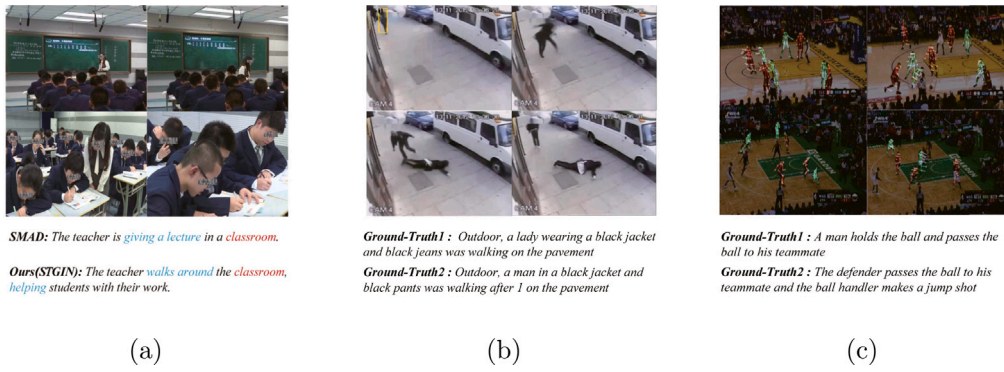


Fig. 1. Examples of video descriptions for classroom instruction, video surveillance and basketball games.

and provide instant detailed, and accurate behavior descriptions (Jain et al., 2022). Benefiting from these advantages, video captioning technology has been successfully applied in various fields such as security surveillance and sports event analysis (Shidik et al., 2019; Mkhallati et al., 2023).

In security surveillance, video captioning focuses on suspicious activities and behavior identification in public places, with an emphasis on individual behavior descriptions, as illustrated in Fig. 1b. In sports events, video captioning focuses on the actions and interactions of athletes during gameplay, with particular emphasis on their dynamic interplay, as illustrated in Fig. 1c. Additionally, it may center on entities and specific scenarios. For instance, the EIKA model (Xi et al., 2025) enhances its performance by leveraging explicit game-related knowledge and implicit visual scene knowledge extracted from the training set. It achieves comprehensive video understanding through integrating explicit and implicit knowledge augmentation. In contrast, the classroom environments involving teacher–student interaction, various teaching methods and different teaching aids are more complex. Additionally, the cameras used to capture classroom videos are not fixed, and they often switch rapidly as the teacher moves (Fang et al., 2023), resulting in significant background changes and diminishing the continuity of teacher actions. As shown in Fig. 1a, the captured classroom video involves teacher–student interaction and scene transformation between the frames when the teacher steps down from the podium to guide students. Therefore, the description of teacher behaviors in the classroom environment should not only consider the individual teacher actions but also focus on the interactions between teachers and students and the background changes between frames.

Considering the significant differences between security surveillance, sports events, and educational environments, the direct application of existing video captioning methods to teaching environments does not provide satisfactory results. Most of the existing video captioning methods, especially those based on skeletal point descriptions or local object features, need to incorporate rich contextual information across video frames, which is crucial for accurately capturing the flow of classroom activities. Furthermore, many methods focus on object recognition or motion tracking, which may work well in structured environments like sports or surveillance but need more depth to describe complex interactions and dynamic changes within a classroom setting. Additionally, behavior recognition methods that reduce background interference by localizing the teacher (Gang et al., 2021) still overlook the interactive nature of teacher–student relationships, which are vital for understanding instructional processes. Therefore, the unique nature of classroom environments necessitates a more specialized video captioning method that can effectively describe teaching behaviors by integrating both spatial and temporal contexts (Li et al., 2022).

This paper presents a comprehensive study on teacher behavior description, with the main contributions including the development of a dedicated dataset and a corresponding method for describing

teacher behaviors. We introduce a novel Spatio-Temporal Graph Interaction Network (STGIN) designed to generate detailed behavior descriptions from classroom videos. STGIN comprises two key modules: the teacher–student Spatial Interaction (TSI) module, which constructs spatial interaction graphs to model object interactivity within each key frame, and the Teacher Temporal Context (TTC) module, which captures temporal dependencies across the video to enhance contextual understanding. These modules jointly inform a description generator that produces rich and semantically meaningful sentences by integrating visual and temporal cues. To support this research, we construct a new Teacher Behavior Description (TBD) dataset, consisting of 2000 videos collected from diverse classroom environments, each annotated with multiple fine-grained behavior descriptions. Extensive quantitative and qualitative experiments conducted on the TBD dataset and two additional generic scene datasets demonstrate the effectiveness and generalizability of our proposed approach.

The contributions of this paper are mainly as follows:

- We propose a spatio-temporal graph interaction network (STGIN) for teacher behavior description in classroom videos.
- We design a teacher–student spatial interaction module (TSI) to model the interaction between teachers and students using spatial location information.
- We present a teacher temporal context module (TTC) to capture the temporal relationships of the same object across video frames.
- We construct a teacher-centered video captioning dataset, which is the first large-scale teacher behavior description dataset in a classroom scenario setting.

2. Related works

2.1. Teaching behavior analysis

The teaching behaviors analysis is an educational research and evaluation approach to assess teaching quality, promote improvements, and enhance student learning outcomes (Li and Li, 2021b). As shown in Fig. 2, it aims to systematically observe and analyze teachers' behaviors during the teaching process and conduct relevant analyses. Existing teacher analysis methods can be grouped into two categories: quantitative analysis and qualitative analysis. For quantitative analysis, Hsiao et al. (2022) proposed a classification method for teacher teaching behaviors, recording and analyzing teacher behaviors' frequency and temporal distribution. Based on the behavior patterns between teachers and students in robot-assisted teaching, Kucuk and Sisman (2017) employ quantitative content analysis and Lag-Sequential Analysis (LSA) to analyze teacher–student interactions in one-on-one robot-assisted teaching processes. Zhao et al. (2023) proposed a correlation analysis approach to explore the mechanisms underlying the transformation of teacher behavior, eye movements, and gestures. The qualitative analysis can objectively describe teachers' teaching behaviors, yielding

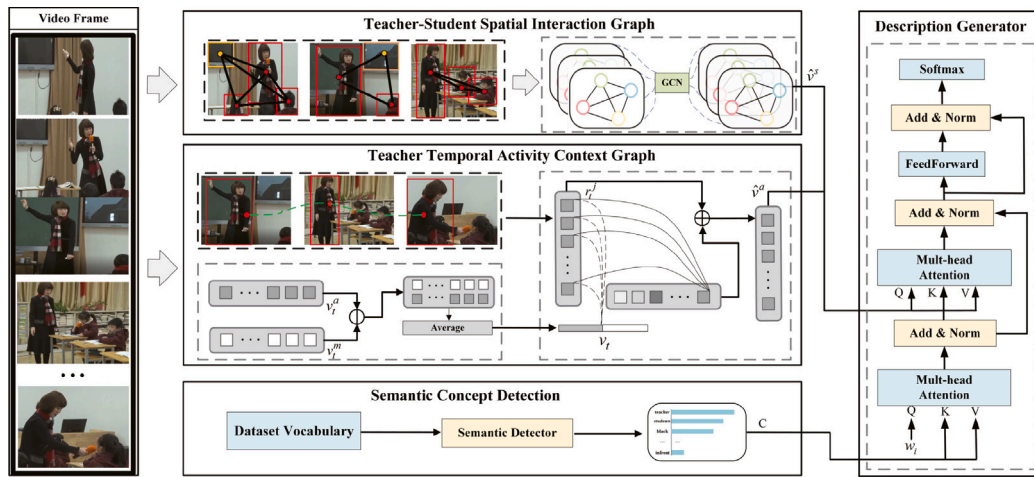


Fig. 2. Architecture of the proposed spatio-temporal graph interaction network.

a deeper understanding of various influencing factors in the teaching process. Rodgers et al. (2019) proposed a multimedia observation tool that allows dedicated personnel to record teachers' teaching actions in real-time. This tool can provide unbiased descriptive data for teachers, facilitating their professional development.

However, such methods require manual identification and encoding, which is time-consuming and labor-intensive. Meanwhile, the data samples are small and insufficient to evaluate teacher behavior accurately. To enlarge data samples, reduce labor costs, and improve the efficiency of teaching behavior analysis, some researchers attempt to utilize machine learning algorithms to analyze and predict teaching behaviors (Gang et al., 2021; Ma et al., 2023). Gang et al. (2021) proposed a teacher-set recognition and extraction algorithm to identify teachers in teaching videos and reduce interference from classroom backgrounds. Ma et al. (2023) constructed a teacher behaviors dataset from real classroom environments and proposed a teacher behavior recognition method based on multi-stream graph convolutional networks to integrate joint, skeletal, and motion information. The above methods, however, focus on statistics and clustering of behaviors and lack objective descriptions of teacher behaviors, which may hinder subsequent teaching evaluations and performance feedback. This paper proposes an automatic generation method for describing teacher behaviors to fill the gap in teacher behavior descriptions.

2.2. Video captioning methods

Video captioning, a critical task in video-language understanding, has evolved from rule-based systems to deep learning frameworks. Early approaches relied on predefined text templates and manual linguistic rules, which were labor-intensive and rigid in describing dynamic scenarios like classroom interactions (Das et al., 2013). The emergence of encoder-decoder architectures, such as the soft attention mechanism by Xu et al. (2015), marked a paradigm shift by enabling context-aware caption generation. Subsequent advancements focused on multimodal fusion: Rahman et al. (2019) integrated audio-visual-textual features, while Video-BERT (Sun et al., 2019) unified modalities via a shared Transformer. Concurrently, researchers enhanced spatiotemporal modeling through object-aware techniques—Bai et al. (2021) embedded frame-level spatiotemporal features, and OA-BTG (Zhang and Peng, 2020) aggregated object relationships via bidirectional time graphs. To improve textual quality, syntax-aware attention (Zheng et al., 2020) and external language models (Zhang et al., 2020) were introduced to refine grammatical coherence and vocabulary coverage. However, these methods often neglected fine-grained object interactions and temporal evolution, limiting their ability to describe complex behaviors like teacher-student dynamics in educational settings.

Recent innovations address these gaps through contrastive learning and tracking-guided architectures. Jin et al. (2024) proposed a Hierarchical Banzhaf Interaction framework, leveraging multivariate cooperative game theory to model fine-grained video-text correspondences and mitigate bias, achieving superior performance in cross-modal tasks. Meanwhile, Luo et al. (2025) introduced Track4Cap, which enhances captioning efficiency by integrating multi-object tracking with object relation encoding. Despite these advances, existing methods still struggle to simultaneously capture global dependencies and discriminative object features in classroom scenes. In contrast, our proposed method can draw global dependencies between teachers and students in classroom scenes while capturing discriminative object features in videos, reflecting interactive behaviors between teachers and students during teaching processes.

3. Proposed method

Fig. 2 illustrates the overall framework of the spatio-temporal graph interaction network (STGIN), which consists of three main modules: teacher-student spatial interaction module (TSI), teacher temporal context module (TTC), and Description Generator. Specifically, in the TSI, the relationships between teachers and students are modeled in the form of a graph, capturing both appearance and positional relationships. Afterward, these representations are learned through graph neural networks. Next, in the TTC, by selecting candidate features of the teacher from each key-frame and using the frame-level features of the video as guidance, we construct a graph-based attention mechanism to model temporal relationships across the entire video sequence. The description generator focuses on temporal and spatial visual features while integrating semantic textual features to generate sentences.

The main contribution of this work lies in the innovative integration of spatio-temporal interaction modeling through graph-based attention mechanisms, allowing for a more comprehensive understanding of both spatial teacher-student interactions and the temporal context of the video. This approach effectively captures dynamic interaction patterns over time, providing a more accurate and context-aware generation of descriptive sentences.

3.1. Teacher-student spatial interaction module

The interaction between teachers and students is an indispensable step in exploring teacher behavior. In this section, we design a teacher-student spatial interaction (TSI) module to build the teacher-student spatial interaction graph. Given a video, an initial step involves applying Yolov5 (Jocher et al., 2022) to each key-frame to predict multiple bounding boxes labeled as humans. Then the features $S^i = \{s_k^i\}$ are

extracted from each bounding box area through RoIAlign (He et al., 2017), where s_k^i represents the k_{th} object feature in i_{th} keyframe. The appearance relationship between two targets is calculated using the pairwise cosine similarity of the visual features of the targets.

$$R_a(x_i^a, x_j^a) = \exp \left(\frac{\cos(x_i^a, x_j^a)}{\max_{i,j} (\cos(x_i^a, x_j^a))} \right). \quad (1)$$

The positional relationship between two targets is represented by the Euclidean distance between the teacher (T) and the student (S). If the captured distance is less than a specific threshold, the positional relationship between these two individuals is considered important.

$$R_s(x_i^s, x_j^s) = I(d(x_i^s, x_j^s) < \mu). \quad (2)$$

Here, I refers to the indicator function, d refers to the Euclidean distance function, and μ is the distance threshold. After that, we define the teacher-student behavioral relationship graph to learn the relationship information from the target object in each frame, which is then used to update the object features. The composite function defines the relationship value:

$$G_{ij} = f(R_a(x_i^a, x_j^a), R_s(x_i^s, x_j^s)), \quad (3)$$

where $R_a(x_i^a, x_j^a)$ denotes the appearance relationship between two participants. The location relationship is computed from $R_s(x_i^s, x_j^s)$. The function f fuses the appearance and location relationships into scalar weights. In our paper, we use the following function to compute the relationship values:

$$G_{ij} = \frac{1}{Z_i} R_s(x_i^s, x_j^s) \cdot g(R_a(x_i^a, x_j^a)), \quad (4)$$

where $g(\cdot)$ represents the non-linear activation function ReLU, Z_i is a normalization factor that ensures the sum of relationship values from node i to all other nodes j is 1.

$$Z_i = \sum_{j=1}^N R_s(x_i^s, x_j^s) \cdot g(R_a(x_i^a, x_j^a)). \quad (5)$$

Subsequently, the constructed graph is iteratively updated to capture and analyze the spatial interaction features between students and the teacher within the frame. The updating process is defined as follows:

$$H^{(l+1)} = \text{ReLU} \left(H^{(l)} + \Lambda^{-\frac{1}{2}} G_l \Lambda^{-\frac{1}{2}} H^{(l)} W^{(l)} \right). \quad (6)$$

Here, $W^{(l)}$ in $\mathbb{R}^{d \times d}$ denotes the weight matrix of the l th layer, and then we obtain the updated H^N , where l is the number of graph convolution layer. And we use a mean-pooling to aggregate the node representations for each frame averagely, denoted as $\hat{v}^s$.

$$\hat{v}^s = \frac{1}{N} \sum_{j=1}^N H_j^l. \quad (7)$$

3.2. Teacher temporal context module

Based on the analysis of teaching videos, it is evident that the teacher's movement information exhibits a distinctively dynamic and interactive nature over time. Their movement patterns vary significantly while lecturing at the podium compared to when they are walking around the classroom or interacting with students. Similarly, we observe that background information is crucial, especially when the teacher is dynamically engaged. This context includes factors such as classroom layout, student activities, and environmental conditions. When the teacher lectures at the podium, the background is a blackboard (or electronic whiteboard). When the teacher walks around the classroom or interacts with students, much background information about the students or the classroom appears.

While recent methods like HTRM-Net (Li et al., 2025) have made advancements in temporal modeling for repetitive action counting, these approaches primarily focus on numerical counting tasks rather than descriptive caption generation. In educational settings, applying robust temporal relation modeling to tasks that require accurate periodicity tracking and semantic context integration, such as repeated student guidance or blackboard writing, remains a significant challenge. In this section, we design a teacher temporal context module to construct the teacher temporal context graph, which utilizes the frame-level features with backgrounds as guidance to capture the temporal paths of significant targets in the video. We first detect possible teacher targets by the target detector. Due to the predominant presence of teachers in the video, we prioritize the selection of the top N bounding boxes with the highest confidence scores to ensure that the most relevant and significant instances of teacher activity are captured for further analysis. In the end, the teacher features $R^j = \{r_t^j\}$ are extracted from each bounding box via RoIAlign. Where r_t^j represents the t_{th} object feature in j_{th} key-frame.

Considering the background information, 2D CNNs and 3D CNNs are employed to extract appearance features $v_t^a \in \mathbb{R}^{D_a}$ and motion features $v_t^m \in \mathbb{R}^{D_m}$ for each key-frame, respectively. Then we build a graph neural network for the entire video sequence in which each object features r_t is seen as a object node. The Integration feature $v_t = [v_t^a \parallel v_t^m] \in \mathbb{R}^{D_f}$ is a guide node to guide the correlation computation and information exchange of object node, where $[\cdot \parallel \cdot]$ denotes concatenation. The correlation score e_{ij} between the i th key-frame and the j th object is calculated as follows:

$$e_{ij} = \sigma \left(W_f [W_i v_t \parallel W_r r_t^j] \right), \quad (8)$$

where W_f , W_i and W_r are learnable parameters, and $\sigma(\cdot)$ denotes the activation function. Then the enhanced object features $\hat{v}^a$ are obtained by weighting each object according to its corresponding associated weights:

$$a_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}, \quad (9)$$

$$\hat{v}^a = W_{\text{attn}} \left(v_t + \alpha_{ij} \hat{r}_t^j W \right). \quad (10)$$

3.3. Description generator

With the aggregation feature $\hat{v}^s$ and $\hat{v}^a$ provided by STGIN, we integrate them into a Transformer-based decoder for sentence generation. Let $w_i = \{w_0, w_1, \dots, w_{T-1}\}$ be the set of words generated up to time step $T-1$. After that, we employ a multi-label classifier known as the Semantic Concept Detection (SCD) (Chen et al., 2020) module to enhance the quality of the generated text descriptions by identifying and highlighting key elements within the video content, thereby facilitating the generation of more accurate and contextually relevant descriptions.

SCD generates a k -dimensional semantic label vector from the global features of the video. Each dimension's value represents the probability that a specific semantic concept is related to the video. These k semantic concepts are selected semi-manually from the training set's vocabulary, focusing on the most frequent and meaningful nouns, verbs, and adjectives. Pennington et al. (2014). Specifically, given the global features v_i of i th video, the corresponding subtitle C and the truth label, we set $\hat{p}_{i,j} = 1$ if the j th label appears in at least one caption C , otherwise $\hat{p}_{i,j} = 0$. A sufficiently large multilayer perceptron (MLP) is utilized to learn semantic representations from the samples. The input feature set is chosen based on experimental results for each dataset. The SCD is trained by minimizing the loss function:

$$\mathcal{L}_S = \frac{1}{N} \sum_{i=0}^{N-1} \sum_{j=0}^{K-1} [\hat{p}_{i,j} \log(q_{i,j}) + (1 - \hat{p}_{i,j}) \log(1 - q_{i,j})]. \quad (11)$$

We compute the attention between the embedding features of the previously generated words and the semantic keyword embedding features. In this process, w_i serves as the query, while C functions as both the key and value for the initial cross-modal attention, as defined by the following equation:

$$Z_i^l = LN(w_i + MHA(w_i, C, C)). \quad (12)$$

The cross-attention component Z_i aligns the previously generated descriptive words with the semantic keywords, enabling the decoder to emphasize these keywords as it generates the subsequent word, thereby maintaining semantic coherence. In the following layers, the sentence decoder processes each word as input, learning to predict the subsequent word based on $\hat{v}^s$ and $\hat{v}^a$.

$$H_i^l = MHA(Z_i^l, \hat{v}^s, \hat{v}^s) + MHA(Z_i^l, \hat{v}^a, \hat{v}^a), \quad (13)$$

$$Y_i^l = LN(H_i^l + Z_i^l). \quad (14)$$

The feature Y_i^l is processed through a feed-forward and normalization layer to produce the output of the $(l + 1)$ th layer in the decoder. With N layers in total, the output of the final layer, Y_d , is passed to a softmax layer, generating a score vector p_{voc} . The word associated with the highest score is selected as the predicted word for the current time step, and this process repeats until the decoder generates the end-of-sequence token.

$$p_{voc} = softmax(W_{voc}Y_d + b_{voc}), \quad (15)$$

where W_{voc} and b_{voc} are learnable parameters.

The cross-entropy loss function is commonly used to train video captioning models. During training, the correlation between the generated words and the reference captions is calculated as the loss to penalize the model, thereby generating sentences similar to the reference captions. In this case, longer captions generated by the model are more likely to be penalized by the loss function, leading the model to output shorter captions with insufficient semantics or sentence structure. Here, we use the length of the corresponding reference captions to weigh the original cross-entry loss. The model can be trained by minimizing the loss function:

$$\mathcal{L}(\Theta) = -\frac{1}{N} \sum_{i=1}^N \frac{1}{\gamma_i^\beta} \sum_{t=1}^{L_i} (y_{i,t} \log \hat{y}_{i,t} + (1 - y_{i,t}) \log(1 - \hat{y}_{i,t})), \quad (16)$$

where Θ denotes the model parameters, N is the batch size, $\gamma_i \in \mathbb{N}^+$ represents the word count of the reference caption for the i th sample, $\beta \in [0, 1]$ is a tunable hyperparameter governing the brevity-semantics trade-off (lower β strengthens the weight for longer captions), $y_{i,t} \in \{0, 1\}$ indicates the ground-truth word token at time step t , $\hat{y}_{i,t} \in [0, 1]$ corresponds to the predicted word probability, and L_i is the caption length of the i th sample. The inverse weighting term $\gamma_i^{-\beta}$ dynamically scales the loss contribution based on caption length—longer reference captions (larger γ_i) receive reduced penalties to prevent over-suppression of semantically dense descriptions.

4. Experiments

4.1. Datasets

In this section, we provide the TBD subtitle dataset and describe the data collection process and annotations. In the data collection process, we deliberately gathered teaching videos from a broad array of subjects — such as mathematics, English, Chinese, science, physics, history, and geography — across various educational stages, from elementary school to university. This approach was intended to ensure the dataset's diversity and enhance its generalizability. We examined the influence of subject matter, educational level, and class size on the model's performance. While these factors do lead to noticeable variations in teaching styles and classroom dynamics, their effect on overall model

Table 1

Data comparison between the TBD dataset and other datasets.

Dataset	Domain	Videos:clips	Sent	Sent/clip
MSR-VTT (Xu et al., 2016)	Open	7k:10k	200k	20
MSVD (Chen and Dolan, 2011)	Open	2k:2k	70k	35
ActivityNet (Caba Heilbron et al., 2015)	Open	20k:100k	100k	1
YouCook2 (Zhou et al., 2018)	Cooking	2k:15.4k	15.4k	1
TBD	Education	2k:2k	20k	10

performance is relatively limited and remains within an acceptable range. Each video is approximately 10–12 s long and includes 10 descriptive sentences. Unlike conventional behavior recognition datasets that use action labels or bounding boxes for classification, our dataset is designed specifically for behavior description, adopting a video-text pair format to capture semantic and temporal aspects of teacher actions. This annotation format allows for richer narrative descriptions of teaching behavior, rather than simple action identification. We have performed a multi-angle annotation strategy during the dataset construction and conducted inter-annotator agreement checks during the labeling process. Specifically, to ensure fine-grained behavioral descriptions, segments longer than five seconds were annotated by three trained crowdsourced annotators who provided multi-level details, including body posture, spatial relationships, and interaction objects. A dedicated annotation tool facilitated consistency checks and expert reviews, while iterative calibration over 5 months ensured reliability. The selected videos are sourced from the China Education Resources Public Service Platform (<https://www.eduyun.cn>). As shown in Fig. 3, our dataset is particularly challenging due to the presence of multiple students and the frequent teacher–student interactions captured in the videos. These interactions include various teaching behaviors such as questioning, explaining, and moving around the classroom, which add to the complexity and diversity of the dataset. This makes the task of accurately describing teacher behavior even more difficult. As shown in Fig. 4, we conducted a detailed dataset analysis. Each video segment is approximately 10–12 s long, with a total of 2000 videos. The dataset covers various subjects, including math, literature, English, and other courses, high-lighting its diversity. These videos are sourced from various educational levels such as elementary school, middle school, high school, and college, further demonstrating the comprehensive nature of the dataset. Additionally, the sentence length in text descriptions primarily ranges from 10 to 20 words, which is longer than typical datasets, adding to the complexity and challenge. The frequent use of verbs such as “stand”, “use”, “suggest”, and “point” in the descriptions reflects common actions in classroom settings, providing a rich annotation that captures various teaching behaviors and interactions.

The statistical analysis in Table 1 quantitatively compares the TBD dataset with mainstream video captioning benchmarks (MSR-VTT (Xu et al., 2016), MSVD (Chen and Dolan, 2011), ActivityNet (Caba Heilbron et al., 2015), YouCook2 (Zhou et al., 2018)), particularly through key metrics including captions per video clip and total sentence quantity. The MSR-VTT dataset serves as a significant benchmark for video captioning, comprising 10,000 video clips, with each clip annotated by 20 unique English descriptions. These videos span 20 diverse categories, including music, sports, and movies. The MSVD dataset comprises 1970 short video clips sourced from YouTube, each annotated with multilingual sentences. For our experiments, we focus on the English captions, which total approximately 40 per video. The ActivityNet dataset features 20,000 untrimmed YouTube videos, each averaging 120 s in length. This dataset provides 100,000 caption annotations, with most videos containing over three annotated events. YouCook2 is the largest instructional video dataset focused on task-oriented cooking activities within the vision community. It comprises 2000 long, untrimmed videos covering 89 different cooking recipes, with an average of 22 videos per recipe.

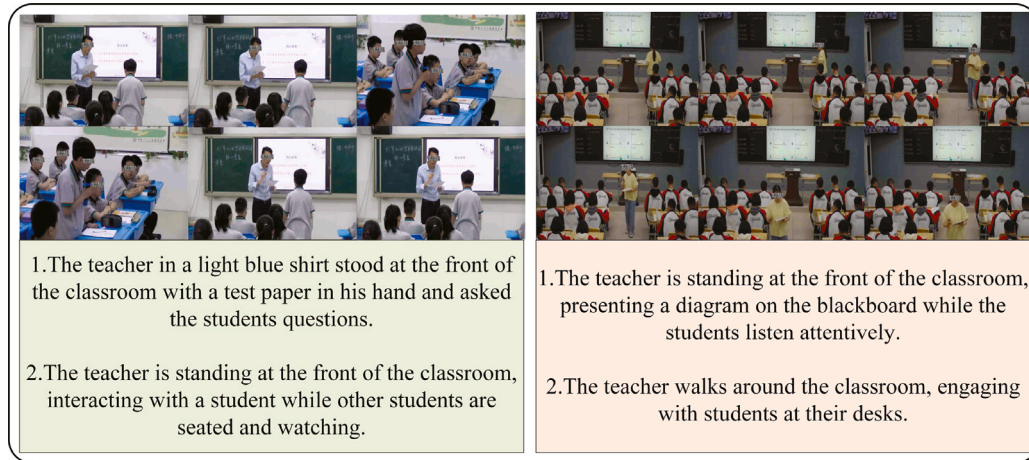


Fig. 3. Examples of TBD datasets.

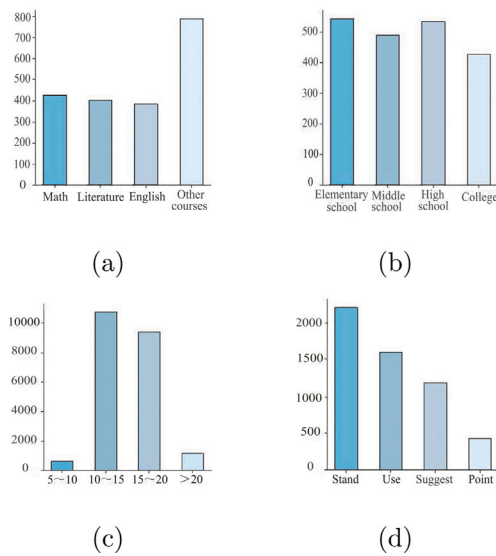


Fig. 4. Statistical Results of (a) Section Distribution, (b) Course Distribution, (c) Description Length, and (d) Verb Word Frequency on the TBD Dataset.

Table 2

Comparison results with different datasets. Video similarity, description length, sentence variance, and syntactic complexity in video analytics metrics represent mean inter-frame visual similarity, speech rate measured through words per second, statistical dispersion of caption sentence lengths, and average hierarchical depth of syntactic parse trees.

Dataset	Video similarity	Description length(words/sen)	Sentence variance	Syntactic Complexity
MSR-VTT	0.64	9	4.22	7.52
MSVD	0.77	8	2.86	6.38
ActivityNet	0.61	13.5	6.10	8.64
YouCook2	0.58	12	4.58	6.43
TBD	0.72	15.55	11.58	10.95

The TBD dataset demonstrates significant advantages in several key aspects compared to other benchmarks. As shown in Table 2, the Central Similarity Quantization (CSQ) method (Yuan et al., 2020) is applied to enable efficient global similarity computation for video segments within this dataset. Statistical analysis reveals that the TBD dataset achieves the longest average description length, the highest sentence-length variance, and relatively high syntactic complexity among existing benchmarks, with the latter metric quantified by syntax tree depth following the methodology of Chen and Manning (Chen and Manning,

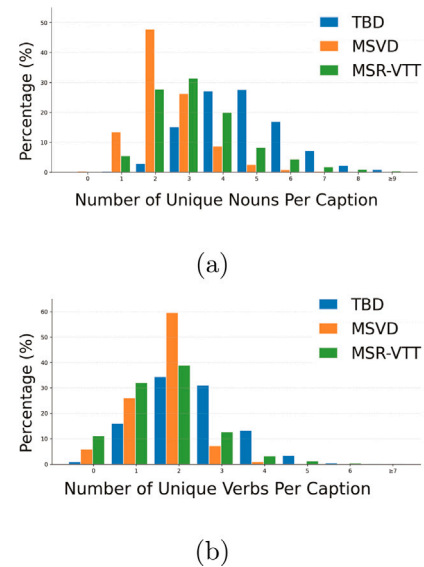


Fig. 5. Distribution of unique words per caption, (a) number of unique nouns per caption, (b) number of unique verbs per caption.

2014). These metrics indicate that the TBD dataset not only includes more detailed and varied descriptions but also employs more complex grammatical structures, which poses a greater challenge for models.

Fig. 5 presents the distribution of unique words per caption. The distinct patterns in noun and verb usage shown in Fig. 5a and Fig. 5b demonstrate that while the TBD dataset is domain-specific, it maintains significant lexical diversity compared to open-domain datasets. This is evidenced by a substantial number of unique vocabulary items and comparatively lower repetition frequencies, even within the constrained context of classroom environments where visual elements show limited variation. In addition to these quantitative indicators, the TBD dataset provides detailed descriptions of various teacher behaviors and states across a wide range of real classroom scenarios. The application of the TBD dataset in the future will greatly enhance the understanding and evaluation of teaching methods, thereby supporting the professional development of educators.

4.2. Implementation details

The TBD dataset is randomly divided into training, validation, and test sets, containing 1200 videos, 400 videos, and 400 videos,

Table 3
Performance comparison with other state-of-the-art methods on the TBD dataset.

Method	Bleu_1	Bleu_2	Bleu_3	Bleu_4	METEOR	ROUGE_L
SMAD (Chen et al., 2020)	77.49	58.01	40.56	28.83	16.74	38.96
VASTA (Ghaderi et al., 2022)	78.11	57.36	39.48	27.72	16.04	37.16
D-LSG (Bai et al., 2021)	77.04	57.25	41.92	28.47	15.34	37.35
V2T-Action (Hendria et al., 2023)	78.21	60.04	40.84	28.63	17.35	40.39
CLIP-DCD (Yang et al., 2022)	74.82	59.61	42.20	30.27	17.77	40.68
CoCap (Shen et al., 2023)	80.40	62.29	42.65	29.91	18.24	39.71
HBI (Jin et al., 2024)	84.52	60.27	44.21	33.45	18.22	41.41
Tracker4Cap (Luo et al., 2025)	73.47	54.92	30.60	27.34	17.95	37.77
STGIN	81.27	62.54	43.15	31.68	18.44	41.85

respectively. To evaluate the captioning performance of our approach, BLEU_n (Papineni et al., 2002), METEOR (Denkowski and Lavie, 2014), ROUGE_L (Chin-Yew, 2004), and CIDEr (Vedantam et al., 2015) are mainly utilized to show the comparison results. BLEU_n measures the similarity between generated captions and reference captions. In this experiment, four NVIDIA GeForce RTX 3090 graphics cards (24 GB video memory/card) were used for model training, and distributed training was conducted through PyTorch. All training processes were completed based on the Ubuntu 20.04 system environment. The choice of parameter n refers to another phrase in the reference captions. The smaller the value of n , the higher the accuracy of generated captions, while the larger the value of n , the higher the semantic similarity between generated captions and references. In contrast, METEOR uses a weighted harmonic mean to evaluate performance. ROUGE measures similarity at the caption level using the longest common subsequences between generated captions and reference captions. CIDEr adopts a voting-based approach, which is more robust against incorrect annotations. For each video clip, 26 frames are selected at uniform intervals. The ResNet-101 (He et al., 2016) and C3D (Tran et al., 2015) are used for feature extraction for 2D CNN and 3D CNN networks, respectively. The 2D CNN is pre-trained on the ImageNet dataset, while C3D is pre-trained on the Sports 1M dataset. We resize each video frame to match the input dimensions of these networks. For 3D CNN, we take 16-frame clips as input. YOLOv5 is used to detect all the objects in the image where the number of bounding boxes N is limited to 5 in each frame. If the number of objects detected in a frame is greater than N , the top N objects with the highest confidence scores remain, while the objects with confidence scores below 0.5 are filtered out.

All reference title sentences are marked and converted to lowercase. After removing punctuation, we collected 3998 word tokens for the TBD dataset and built a vocabulary with words that appear at least twice. For training video-sentence pairs, annotations with fewer than 26 words are padded with zeros, and annotations longer than 26 words are trimmed. During training, the start-of-sentence and end-of-sentence tags are added at the beginning and the end of the sentence, respectively. Words that are not found in the vocabulary are set as unknown labels. Each word is one-hot encoded and embedded into a 512-dimensional space. We set the number of layers in the decoder to 6. In the multi-head attention block of the model, the number of heads is set to 10. For all graph operations, the feature size is set to 1024. In addition, for the semantic concept detector, we semi-manually constructed candidate labels from the training set of each dataset from the MSVD dataset with a few labels of 300, MSR-VTT of 400, and TBD of 300, which includes the most common adjectives, nouns, and verbs.

During the training stage, all parameters are initialized randomly, and the captioning model is optimized using the Adam algorithm. The learning rate is fixed at 1×10^{-6} , and the batch size is configured at 64. During the inference stage, we employ beam search to select the top- k best sentences at each time step, which are used as candidate sentences to generate new top- k best sentences at the next time step. The beam size k is set to 6, and the best checkpoint is selected based on the BLEU₄ score.

4.3. Performance comparison on TBD dataset

4.3.1. Quantitative evaluation

Our proposed model was compared with the state-of-the-art methods on the TBD, MSVD, and MSR-VTT datasets. Table 3 shows the comparison results on the TBD dataset. Our method outperforms all compared methods on all evaluation metrics, which validates the effectiveness of our proposed STGIN. In the compared methods, first, models based on semantic assistance are evaluated. The Semantics-Assisted method proposes a metric to compare the quality of semantic features, generating meaningful semantic features for videos. However, this method achieves a lower BLEU₄ score of 28.83 compared to our method's 31.68, due to its reliance on semantic features that may not effectively capture the dynamic interactions in classroom scenes. VASTA (Ghaderi et al., 2022) uses visually based semantic context vectors obtained from all the captions of each sample to provide high-quality semantic guidance for the decoder. Despite this, it scores 27.72 on BLEU₄, significantly lower than our method, as it struggles with the diverse and complex interactions typical in classroom environments. D-LSG (Bai et al., 2021) introduces a new conditional graph that fuses spatio-temporal information into potential target suggestions, which are summarized as dynamic visual words. While innovative, it scores 28.47 on BLEU₄. Our method surpasses this by 11%, highlighting the importance of more robust modeling of target relationships, which D-LSG's approach may lack. V2T-Action (Hendria et al., 2023) merges action features as edge features into graph neural networks, scoring 28.63 on BLEU₄. Although it attempts to integrate action features, it still falls short compared to our method due to the limited scope of its action representations in capturing nuanced teacher-student interactions in classrooms. Compared to CLIP-DCD (Yang et al., 2022), our model improves by 4.7% and 2.9% in BLEU₄ and ROUGE_L, respectively. The improvement is due to that the CLIP-DCD focus on feature extraction methods, which ignores object modeling.

HBI (Jin et al., 2024), while achieving the highest BLEU₄ score of 33.45, primarily leverages a hierarchical game-theoretic approach to model fine-grained semantic interactions. It introduces Hierarchical Banzhaf Interaction to simulate the correspondence between video clips and textual words and reconstructs representations by fusing single-modal and cross-modal components. While effective, its coarse-to-fine semantic modeling may generalize better in broader domains, but in the classroom-specific context of TBD, STGIN achieves a better trade-off between fine-grained temporal alignment and visual-semantic relation modeling. Our STGIN model performs competitively, trailing HBI by only 1.77 points in BLEU₄ score. However, it outperforms HBI in terms of METEOR, with a score of 18.44 compared to 18.22, and also leads in ROUGE_L, achieving 41.85 against HBI's 41.41. Tracker4Cap (Luo et al., 2025), on the other hand, advocates for frame-level features using a Tracking-guided Information Augmentation strategy. It integrates object tracking via the Frame-by-Frame Multi-object Tracking module and enhances inter-object relation modeling through the Object Relation Encoder. Despite its compact and efficient design, it performs significantly lower across all evaluation metrics, with a BLEU₄ score of only 27.34. This is largely due to its relatively shallow exploitation of temporal dynamics and lack of deeper semantic integration across frames, which limits its effectiveness in capturing the rich interactions and narrative structure present in classroom scenes.

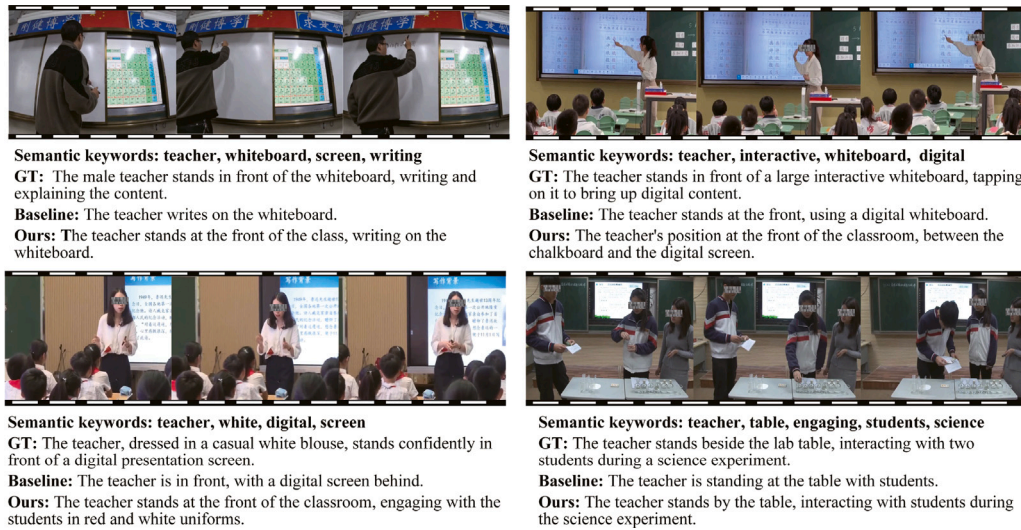


Fig. 6. Qualitative results of four videos from the TBD datasets. The first line in each example is one of the ground truth captions and the second line is generated by our method.

4.3.2. Qualitative evaluation

To visualize the comparative advantage of the proposed method, Fig. 6 juxtaposes sample keyframes and their corresponding generated descriptions from our approach against those of the SMAD baseline model on the TBD dataset. The keyframes illustrate various actions typical of a real classroom setting. Compared to the baseline method, our approach consistently provides more detailed and contextually accurate descriptions. For instance, while the baseline generically describes a teacher standing in front of a screen, our method specifies the teacher's actions and interactions within the scene, such as pointing to a diagram or engaging with students. This enhanced descriptive capability demonstrates the effectiveness of our approach in capturing and conveying meaningful information from classroom video content. In addition, the figure shows the semantic keywords detected by the semantic concept detection method. We can see that it effectively includes semantic keywords in the description, which demonstrates the usefulness of combining our semantic features into the decoder.

These generated descriptions can be used to provide teachers with structured feedback on their teaching practices. For example, teachers can watch the entire teacher behavior analysis of the recorded class provided by the system after class. In the detailed behavior description, the system can automatically highlight the parts where teachers actively interact with students, rather than the parts dominated by one-way teaching. For example, in a specific case, our model accurately identified the teacher guiding the group discussion, capturing not only the teacher's movements between groups but also the students' participation through gestures and reactions. This detailed feedback enables teachers to reflect on the balance among different teaching strategies and students' responses to these strategies.

4.4. Ablation studies

The proposed TSGIN framework consists of three key components: the teacher-student spatial interaction module (TSI), the teacher temporal context module (TTC), and the Description Generator (DG). To demonstrate the effectiveness of each component, we conducted a quantitative analysis, with the results summarized in Table 3. The #0 variant is the baseline model, where all proposed modules are removed: the feature maps obtained from both CNNs and object detection methods are spatially mean-pooled into vectors and fed into a transformer decoder.

As shown in Table 4, the addition of TSI, TTC, and DG modules each significantly enhances the performance of the baseline model, indicating the utility of these components. Specifically, incorporating the TSI

Table 4

The effectiveness of the different components of our approach.

#	TSI	TTC	DG	Bleu_4	METEOR	ROUGE_L
0	✗	✗	✗	27.42	15.24	38.06
1	✓	✗	✗	28.63	16.77	40.25
2	✓	✓	✗	30.56	18.11	41.34
3	✓	✓	✓	31.68	18.44	41.85

Table 5

Ablation study of different object detection methods on TBD dataset.

MODEL	Bleu_4	METEOR	ROUGE_L
FASTER R-CNN	31.10	17.97	41.38
SSD	31.52	17.25	40.78
YOLOV5	31.68	18.44	41.85

module yields the highest performance gain, due to its ability to capture richer spatial information, thereby improving the model's recognition and understanding capabilities. The TSI module's ability to capture spatial behavior interactions is particularly critical, underscoring the importance of spatial information in understanding teacher-student interactions. The combined use of TSI, TTC, and DG modules produces the best outcomes, highlighting the complementary benefits of spatial, temporal, and semantic features in enhancing the model's capability to analyze and describe classroom behaviors effectively.

We compare the proposed method under different conditions. The beam size is controlled in the range of 1, 3, 5, 6, 8, 10. The model performance variation with parameters is shown in Fig. 7. The setting of the beam size K has a large impact on the model. When K is less than 6, BLEU_1, METEOR, and ROUGE_L all increase gradually, which indicates that STGN's ability to capture image information is gradually increasing. The generated subtitles are further matched with the real subtitles gradually. It is not until after K reaches 6 that the indexes begin to gradually decrease. Therefore, it is more appropriate to set the beam size to 6. For the object feature extraction method, we use three different object detection methods for the ablation study, i.e., SSD (Liu et al., 2016), Faster R-CNN (Ren et al., 2015), and YOLOv5. As shown in Table 5, among three object detection methods, YOLOv5 performs best on the TBD dataset, followed by Faster R-CNN, and SSD. The more accurate the object detection, the more precise the generated description. Conversely, inaccurate bounding boxes introduce more noise, affecting feature extraction. Therefore, target-based graph methods rely on high-precision object detection models to ensure effective feature extraction.

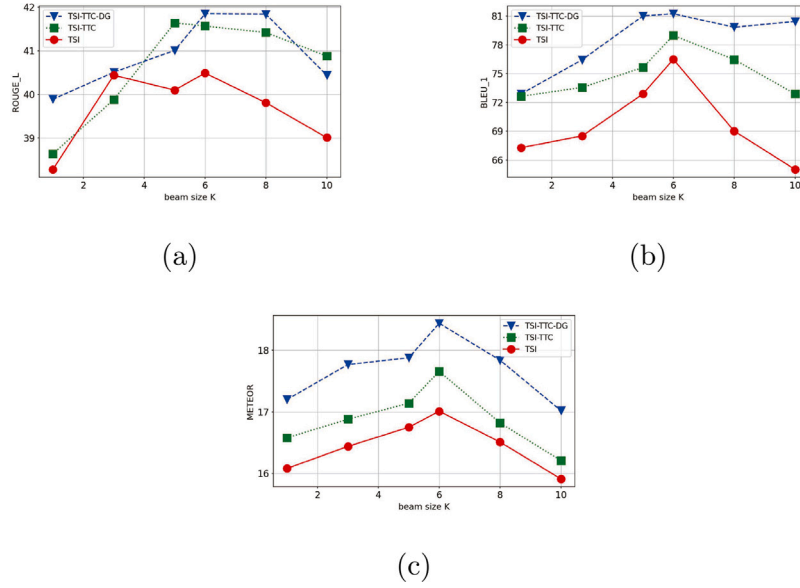


Fig. 7. Effect of different beam sizes on model performance in experiments with the TBD dataset.

Table 6

Effect of different parameter β on the results of TBD dataset.

β	Bleu_4	METEOR	ROUGE_L
$\beta = 0.0$	31.02	18.69	41.65
$\beta = 0.2$	30.82	18.13	40.60
$\beta = 0.4$	31.68	18.44	41.85
$\beta = 0.6$	29.52	17.80	41.29
$\beta = 1.0$	30.35	17.86	40.97

Finally, We conducted ablation experiments for the hyperparameter β in the loss function, $\beta \in [0, 1]$, which reflects the balance between brevity and precision of the output headings. When $\beta = 0$, the loss function reduces to the standard cross-entropy loss, which could result in the model producing shorter headlines. On the other hand, when $\beta = 1$, the optimizer may struggle to differentiate between the words in the generated headlines, potentially leading to the repetition of words. In Table 6, the proposed can achieve better results when β is set to 0.4.

4.5. Generalization and robustness analysis

To verify the generalizability of our STGIN to other descriptive tasks and its robustness across different datasets, we conducted further experimental validation. We compare it with state-of-the-art models on the MSVD and MSR-VTT datasets. As shown in Table 7, our approach outperforms most of the state-of-the-art approaches with various backbones in two datasets, especially on the BLEU metrics. SAAT focuses on the action relationships between targets while referencing the topic and video dynamics to explicitly learn the actions. Despite this, our method shows a notable improvement in the BLEU_4 score on the MSR-VTT dataset, achieving 44.8 compared to SAAT's 40.5, indicating a gain of 4.3 points. This demonstrates that our approach more effectively captures the relevant actions and interactions within the video content. We further compare a few approaches that use object features. ORG-TRL (Zhang et al., 2020) proposes an encoder based on object-relationship graphs that captures more detailed interaction features to enrich the visual representation. OA-BTG employs a bi-directional temporal graph to characterize objects and their relationships. Despite these sophisticated methods, our model achieves higher scores in several metrics, such as a BLEU_4 score of 61.0 on the MSVD dataset, which is significantly higher than ORG-TRL's 55.8 and OA-BTG's 56.9. This improvement underscores the efficacy of our method in leveraging object features to enhance video captioning performance.

On the MSVD dataset, MGRMP (Chen and Jiang, 2021) does not employ object features but better extracts different spatial features through a cyclic region attention module, establishing higher-order relationships across different regions of the frame. Nevertheless, our approach surpasses MGRMP's performance, with our CIDEr score being 105.4 compared to MGRMP's 98.5, indicating an advantage of 6.9. Additionally, HMN (Ye et al., 2022) makes full use of language semantics, it lacks exploration of the relationships between objects. Therefore, our method achieves superior performance on the METEOR and ROUGE_L metrics. While CoCap (Shen et al., 2023) learns subtitles from compressed videos and performed best in METEOR and ROUGE_L scores. Our method also achieves sub-optimal results on this dataset.

5. Conclusion

This paper proposes a novel graph-based method to elucidate the relationships between target features within a classroom environment. By constructing spatial and temporal graphs, our approach captures the intricate dynamics of classroom interactions and enhances the descriptive capabilities of generated sentences. We have built a comprehensive video caption dataset based on real classroom videos, which includes diverse scenarios and teaching behaviors. Meanwhile, semantic concept detection methods are used to detect frame features to obtain the video's most relevant semantic feature representation to reduce redundant information in the video description, making the generated description more concise and efficient. The experimental results demonstrate the effectiveness of the proposed module, highlighting its ability to accurately describe teacher behaviors and interactions in classroom settings.

In the future, we can further improve our methods through the integration of knowledge graphs or large-scale models. These advancements will not only enhance the accuracy and adaptability of our system but also allow it to tackle more complex and nuanced classroom scenarios, ultimately providing deeper and more actionable insights into teaching practices. To support these improvements, we plan to expand the TBD dataset by creating sub-datasets categorized by subject area and educational stage (e.g., primary, secondary). This stratification will enable the design of targeted experiments that consider the distinct characteristics and pedagogical goals of different disciplines and grade levels. Additionally, we aim to incorporate densely annotated subtitles with detailed temporal alignment. This will make it possible to perform fine-grained temporal analyses of teacher behaviors, such

Table 7
Performance comparison with other state-of-the-art methods on MSR-VTT and MSVD datasets.

Method	MSR-VTT				MSVD			
	BLEU_4	METEOR	ROUGE_L	CIDEr	BLEU_4	METEOR	ROUGE_L	CIDEr
SMAD (Chen et al., 2020)	45.8	29.3	63.6	53.2	–	–	–	–
VASTA (Ghaderi et al., 2022)	43.5	29.8	62.3	53.6	56.1	39.1	74.5	106.3
D-LSG (Bai et al., 2021)	44.6	28.8	62.3	51.2	60.9	37.6	75.2	100.8
MGRMP (Chen and Jiang, 2021)	41.7	28.9	62.1	51.4	55.8	36.9	74.5	98.5
ORG-TRL (Zhang et al., 2020)	43.6	28.8	62.1	50.9	54.3	36.4	73.9	95.2
SAAT (Zheng et al., 2020)	40.5	28.2	60.9	49.1	46.5	33.5	69.4	81.0
OA-BTG (Zhang and Peng, 2020)	41.4	28.2	–	46.9	56.9	36.2	–	90.6
HMN (Ye et al., 2022)	43.5	29.0	62.7	51.5	59.2	37.7	75.1	100.2
CoCap (Shen et al., 2023)	44.4	30.3	63.4	57.2	60.1	41.4	78.2	121.5
STGIN	44.8	30.5	62.2	54.8	61.0	40.9	76.8	105.4

as instructional pacing, questioning strategies, and student interaction patterns, offering a dynamic and time-sensitive understanding of teaching processes.

CRedit authorship contribution statement

Yu Xiong: Writing – review & editing, Funding acquisition.
Chengyang He: Writing – original draft, Methodology. **Lulu Chen:** Writing – review & editing. **Ting Cai:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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